

Lab2

Setup

Before anything is started, we are going to load the libraries for the biostatistics/Clinical trials data set

```
library(survival)
data(pbc)
```

Non trivial analysis

Now, we have our data points and we want to do something with it. Lets focus on cholestrol. Lets see cholestrols mean, low, medium, and high values

```
truechol <- pbc$chol[!is.na(pbc$chol)]
summary(truechol)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  120.0   249.5   309.5   369.5   400.0  1775.0
```

Now, we are to add the different levels of cholestrol based on risk.

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##      filter, lag
##
## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
```

```
chollevels <- as.data.frame(truechol)
Chollevels <- chollevels %>%
  mutate(risklevels = case_when(
    truechol < 200 ~ "Low risk",
    truechol < 300 ~ "Medium risk",
    truechol < 400 ~ "High risk",
    truechol > 500 ~ "Severe risk",
  ))
```

Visualization

Here, we will visualize some data regarding the chol levels

```
countsChol <- table(Chollevels)
barplot(countsChol,
  main = "Number of patients per cholestrol risk group",
  xlab = "Category of risk level",
```

```
ylab= "Amount of patients",  
col = "green")
```

